30. Bulimia An article in the *British Journal of Clinical Psychology* indicates that self-esteem was an important predictor of those who will benefit from treatment for bulimia.[4] The table gives the mean and standard deviation of self-esteem scores prior to treatment, at posttreatment, and during a follow-up:

	Pretreatment	Posttreatment	Follow-up
Sample Mean $\bar{x}$	20.3	26.6	27.7
Standard Deviation s	5.0	7.4	8.2
Sample Size n	21	21	20

a. Use a test of hypothesis to determine whether there is sufficient evidence to conclude that the true pretreatment mean is less than 25.

b. Construct a 95% confidence interval for the true posttreatment mean.

c. In Section 10.3, we will introduce small-sample techniques for making inferences about the difference between two population means. Without a formal statistical test, what might you conclude about the differences among the three sampled population means represented by the results in the table?

b. $df = 21 - 1 = 20$

$t_{20;0.25} = 2.086$

$CI = \bar{X} \pm t_{20;0.25} \frac{s}{\sqrt{n}}$

$= 26.6 \pm 2.086 \cdot \frac{7.4}{\sqrt{21}}$

$= 26.6 \pm 3.368$

$CI = (23.232, 29.968)$

c. there's a noticeable increase in the sample mean $\Rightarrow$ positive effect on self-esteem.

a. $H_0: \mu = 25$, $H_a: \mu < 25$ $df = 21 - 1 = 20$

$t^* = \frac{20.3 - 25}{\frac{5}{\sqrt{21}}} = -4.308$

$p_t < 0.05 \Rightarrow$ reject H_0

35. Sodium Chloride Measurements of water intake,